Imitated Detectors: Stealing Knowledge of Black-box Object Detectors

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ABSTRACT

Deep neural networks have shown great potential in many practical applications, yet their knowledge is at the risk of being stolen via exposed services (*e.g.*, APIs). In contrast to the commonly-studied classification model extraction, there exist no studies on the more challenging object detection task due to the sufficiency and efficiency of problem domain data collection. In this paper, we for the first time reveal that black-box victim object detectors can be easily replicated without knowing the model structure and training data. In particular, we treat it as black-box knowledge distillation and propose a teacher-student framework named Imitated Detector to transfer the knowledge of the victim model to the imitated model. To accelerate the problem domain data construction, we extend the problem domain dataset by generating synthetic images, where we apply the text-image generation process and provide short text inputs consisting of object categories and natural scenes; to promote the feedback information, we aim to fully mine the latent knowledge of the victim model by introducing an iterative adversarial attack strategy, where we feed victim models with transferable adversarial examples making victim provide diversified predictions with more information. Extensive experiments on multiple datasets in different settings demonstrate that our approach achieves the highest model extraction accuracy and outperforms other model stealing methods by large margins in the problem domain dataset. Our codes can be found at https://github.com/LiangSiyuan21/Imitated-Detectors.

CCS CONCEPTS

• **Computing methodologies** → **Object detection**.

KEYWORDS

Object detection, model stealing, knowledge distillation

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1 INTRODUCTION

Deep learning techniques often require large amounts of training data to achieve remarkable performance on specific tasks [12, 24, 28, 51]. However, the data collection and model training process are often highly time-consuming and labor-intensive, which makes these deep learning models valuable intellectual properties. Thus, big companies only provide users with API services in order to monetize these deployed deep models. Despite of this, the knowledge of deep models is still at the risk of being stolen [39]. Adversaries could simply perform the model extraction attacks by sending inputs to the model and observing the model predictions without the necessity of accessing the model internals, which may cause severe security problems [17].

In the current literature, deep learning techniques have been proved to be fragile [3, 3, 37, 38, 42, 43] and a long line of studies has been devoted to performing model extraction attacks on the classification task [6, 26, 34, 36, 50]. For object detection [11, 19, 31], another essential yet more challenging branch in computer vision,

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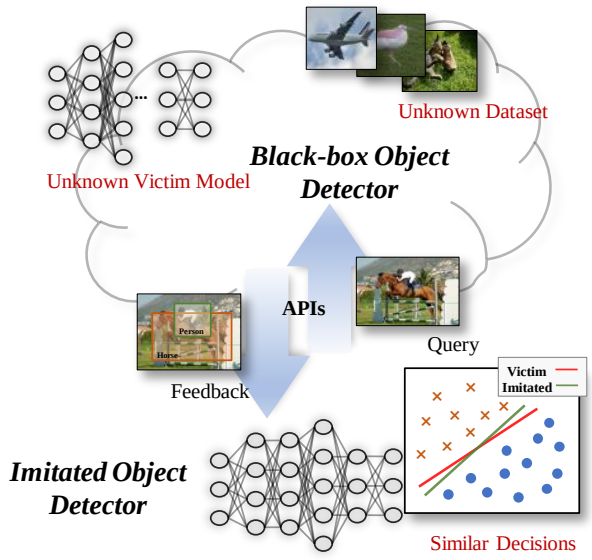


Figure 1: The illustration of black-box knowledge stealing on object detection. In general, the adversary does not know the training data and the victim model in the black-box setting. The adversary could only get the object's category and location by querying the APIs of the black-box victim. Based on this feedback, the adversary would train an imitated object detector with similar decisions to the victim model.

relatively few attempts have been devoted to stealing the knowledge of detectors. Compared to image classification, stealing an object detector primarily faces the following dilemmas due to the high-demanding of the input: (1) Efficiency of problem domain data construction. In contrast to classifiers that would always output predictions for any unseen classes, detectors would recognize unseen objects as background with no predicted locations provided. Thus, collecting and building query data for detectors requires including high-quality objects in the specific problem domain, therefore is more labor-intensive. (2) Sufficiency of model feedback. Given a rational query, it is still non-trivial to train the imitated model based solely on the simple victim model outputs. Compared to the logistic scores returned by classifiers, object detectors often return predicted locations and corresponding hard labels, which may deliver less information for adversaries to learn. Moreover, simply extending existing model extraction attacks from image classification to object detection is highly non-trivial owing to the significant scenario gaps.

In this paper, we for the first time explore the possibility of knowledge stealing on the object detection task. To address the above challenges, we treat the object detection knowledge stealing as a black-box knowledge distillation problem and propose a teacher-student framework for imitating the object detector, deemed as Imitated Detector. **To accelerate the problem domain dataset construction**, we construct the synthetic images to expand the problem domain dataset by the text-image generation process. We first provide the short text sentence inputs with object categories and natural scenes to generate the domain problem objects patches, then we randomly select and place the object patches to form the synthetic images. **To improve the feedback information**, we fully excavate the latent knowledge of the victim models by introducing an additional object detector to generate transferable adversarial examples for natural images [7, 21, 22, 45] These imperceptible perturbations could make the victim model generate more diversified predictions as augmentation pseudo-annotations for the same data, which could deliver more information to adversaries for better training. Extensive experiments on multiple datasets in different settings demonstrate that our approach achieves the highest model extraction accuracy and outperformed other model stealing methods. In addition, we also instantiate that our approach is also feasible for stealing object detectors with different model architectures. In summary, our contributions are as follows:

- We regard knowledge stealing on object detection as black-box knowledge distillation and propose the teacher-student framework to address the problem. As far as we know, we are the first to steal knowledge on object detectors.
- Our Imitated Detector approach introduces text-image generation and transferable adversarial attacks to simultaneously solve the efficiency and sufficiency challenges of training data for object detection knowledge stealing.
- Extensive experiments on multiple datasets in different settings demonstrate that our approach achieves the highest model extraction accuracy and outperforms other model stealing methods.

2 RELATED WORK

2.1 Object Detectors based on Deep Models

The development of object detectors based on deep models can be divided into two stages according to the development of anchors: anchor-based detectors and anchor-free detectors. Object detection usually classifies and regresses candidate regions, which we call anchors. In one-stage detectors, such as YOLO [30], SSD [25], anchors are mainly generated by sliding windows; in two-stage detectors, such as Faster R-CNN [31], Mask R-CNN [14], anchors are proposals generated by the region proposal network. In recent years, anchor-free based detectors have gradually become a popular trend, such as FCOS [33], Free Anchor [48], and FSAF [49]. Since the anchor-free based algorithm does not need to debug the anchor parameters, it is friendly to the constructed dataset. To eliminate the effects of different training data distributions, we use anchor-free based detectors for knowledge stealing.

2.2 Model Stealing in Classified Model

In classified models, the model stealing methods are mainly divided into data-based and data-free. In data-based model stealing, the attacker leverages the classification dataset to query the victim model and train the imitated model using pseudo-labels. Copycat [6] discusses the impact of naturally random images out of the problem domain and a specific problem domain. KnockoffNets [27] can produce nearly 97% accuracy with the victim model under a suitable query dataset. However, an unsuitable dataset will affect

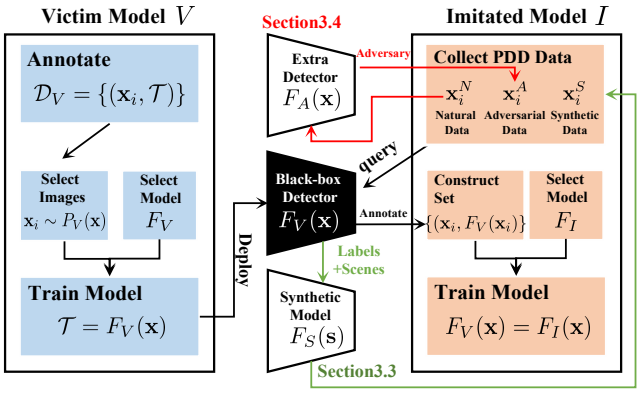


Figure 2: Basic steps for object detection knowledge stealing. The construction of the query dataset and the selection of the imitated model will affect the final accuracy of the model.

the performance of the imitated model. In the data-free setting, attackers cannot use any data. [34] uses random noise data to steal classified models. MAZE [15] feeds generation data from Gaussian Noise to the victim and imitated models, optimizing the generative model using Zeroth-Order Gradient. In object detection, feeding noisy data into the victim model causes the detector to produce no outputs. Therefore, using data-based settings is more suitable for the knowledge stealing of object detectors.

2.3 Transfer-based Adversarial Attack

Black-box adversarial attacks restrict the attacker's access to resources, divided into query-based [1, 2, 4] and transfer-based [23, 35, 40, 41] attacks. The transfer-based attack mainly generates transferable adversarial examples on the surrogate model to attack the victim model, such as DI-FGSM [45], TI-FGSM [7], and Curls & Whey [32], without querying models. Since the transfer-based adversarial attack does not require access to the victim model, it is more popular in lightweight attack scenarios. Although we use adversarial examples to mine the latent information of the victim detector, constant access to the victim model makes model stealing invisible. Therefore, we prefer to employ the transfer-based adversarial attack that does not require access to the victim model.

3 IMITATED DETECTOR

In this section, we first introduce the problem definition of the object detection knowledge stealing; we then elaborate on our overall Imitated Detector framework; we further illustrate the synthetic image generation and adversarial attack process, which would help to accelerate the problem domain data construction and improve feedback information, respectively; finally, we explain our overall two-stage training strategy that could further improve the imitated model accuracy.

3.1 Problem Definition

Victim Object Detector. The victim detector F_V is the target model we aim to perform knowledge stealing. It is often trained on the hand-annotated close-set dataset $\mathcal{D}_V$. A specific problem domain uniquely determines the object categories C for this dataset. To protect the intellectual property and model security, the victim model is often deployed in the form of APIs and can be regarded as a black-box detector to users.

Training Data Distribution. The training data for the detector needs to label objects' categories and their locations. Objects consider out of the categories C as background. We refer to the training data in the categories C as problem domain data (PDD) and out of the problem domain as OPDD. The differences in the categories bring different distributions between the problem domain and out of the problem domain. We call the data used to train the victim model the original domain data (ODD). As far as we are concerned, the overall categories of PDD and ODD are the same. Due to the difference in the numbers of labeled instances, the data distribution of PDD and ODD will also be different.

Knowledge Stealing for Object Detector. Assuming that the digital image $\mathbf{x}$ has N objects, and the object detector F detects the location $\mathbf{b}_n$ and category c_n for n-th object then outputs final detection results $\mathcal{T} = \{\mathbf{t}_1, ..., \mathbf{t}_n\}$. In Figure 2, given the victim detector F_V, the adversaries aim to obtain the imitated detector F_I with the similar function of the victim detector $F_V(\mathbf{x}) = F_I(\mathbf{x})$. In the knowledge stealing stage, the attacker cannot obtain the information of ODD and can only collect and construct the PDD through the feedback categories C of victim model APIs. This disparity in training data distribution brings significant challenges to knowledge transfer.

Attacker's Unknown. The outputs of the object detector are one-hot labels representing the object categories and their locations. In the knowledge stealing process, we assume that the attacker does not know: (1) the structure and parameters of the victim detector; (2) the training dataset of the victim detector; (3) the logistic score feedback by the victim model.

3.2 Overall Teacher-student Framework

In this paper, we regard the knowledge stealing on object detection as a special form of knowledge distillation between the teacher model (victim model) and the student model (imitated model). In knowledge distillation, the student model has access to the teacher object's internal features and training data annotations as follows:

$$\mathcal{L}_{KD}(\mathbf{x}) = \lambda_1 l(\mathcal{T}_{true}, F_S(\mathbf{x})) + \lambda_2 \sum_{i=1}^{M} \phi(F_i^T(\mathbf{x}), F_i^S(\mathbf{x})), \quad (1)$$

where λ_1 and λ_2 are hyperparameters. $l(,)$ and $\phi(,)$ represent the loss functions of the ground-truth and the outputs of the teacher objector, respectively. F^i denotes the i-th detector's output.

For detection knowledge stealing, the student model can only access the one-hot labels and locations of the teacher model, which can be regarded as the last outputs F_T^M. Moreover, the student model cannot access ground-truth annotations. Thus, the goal of knowledge stealing can be written as:

$$\mathcal{L}_{KS}(\mathbf{x}) = l(F_M^T(\mathbf{x}), F_M^S(\mathbf{x})) = l(F_M^V(\mathbf{x}), F_M^I(\mathbf{x})), \quad (2)$$

where we use the pseudo-labels $F_M^I(\mathbf{x})$ of the victim model as the ground-truth to train the imitated model.

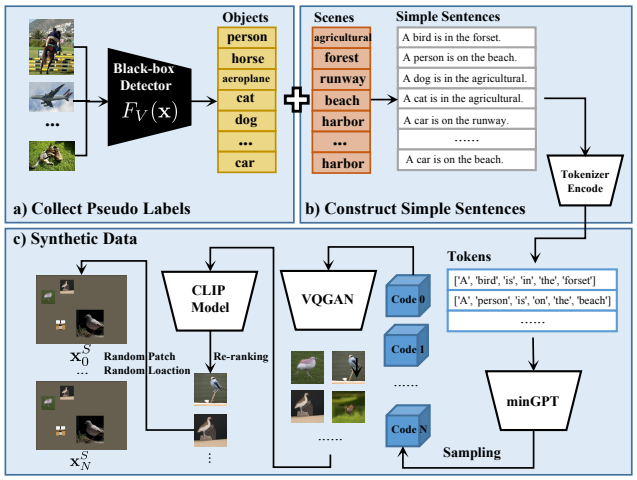


Figure 3: Synthetic data generation process. To save time manually collecting problem domain data, we construct simple text sentences using pseudo-labels generated by black-box models and real-world scene labels. A text-image synthesis model consists of minGPT, VQGAN, and CLIP models. We construct synthetic data in the problem domain.

To obtain an high-accuracy imitated detector, the adversary must construct the problem domain dataset, which is now facing the sufficiency and efficiency challenges of problem domain data collection. As shown in Figure 2, we propose the teacher-student framework named Intimated Detector to address the problems. Regarding the efficiency of problem domain data (query) construction, we introduce the text-image generation process with a trained sentence synthesis model F_G. We first provide the short text sentence inputs with object categories and natural scenes to generate the domain problem objects patches, then we randomly select and place the object patches to form the synthetic images. As for the sufficiency of model feedback information, we employ an adversarial detector F_A and generate adversarial examples based on it. The generated attacks could mislead victim models to provide more diversified predictions. To sum up, the adversary randomly samples the problem domain dataset and queries via the APIs of the victim model to get pseudo-labels, which will be used to train an imitated detector along with the problem domain data.

3.3 Query Construction Acceleration via Synthetic Images Generation

The first step in knowledge stealing is to construct the problem domain dataset. Attackers can collect problem domain data on the Internet-based on the pseudo-label categories returned by the victim detector. Although the attacker does not need to label the instance in the image, it is still challenging to sample a large amount of Internet data. To reduce the time to sift through the problem domain samples manually, we propose a method for structuring synthetic data in the problem domain.

With the rapid development of large multimodal models, large multimodal models perform well in downstream tasks, such as text-image generation, image caption, etc. Since the large multimodal model integrates a large amount of perceptual knowledge, it is possible to utilize the text-image mapping capability of the large model and the high-quality image generation model F_S to generate object patches. We mainly refer to the large model combination of [13], using trained large models such as minGPT, DALL-E, and generation model VQGAN to generate object patches and randomly sample the sizes and positions to construct synthetic data.

In Figure 3, we implement the use of pseudo-label categories and real-world scene categories to form a simple short sentence s. We implement word segmentation in sentence to get the word-token $W(s)$. Next, we sent the word-token $W(s)$ to the encoder and get the encoding code $\hat{\mathbf{z}} = E(W(s)) \in \mathbb{R}^{h\times w\times n_z}$. In VQGAN [8], the generator learns to represent images with codes from a learned codebook $\mathcal{Z} = \{\mathbf{z}_k\}_{k=1}^{K}$. n_z is the the dimention of codes $\hat{\mathbf{z}}$ and $h \times w$ indices the entries in the learned codebook. A subsequent element-wise quantization $q()$ to approximate $\hat{\mathbf{z}}$ to obtain $\mathbf{z}_q$ as follows:

$$\mathbf{z}_q = q(\hat{\mathbf{z}}) = (\arg\min_{\mathbf{z}_k\in\mathcal{Z}} ||\hat{\mathbf{z}}_{ij} - \mathbf{z}_k||) \in \mathbb{R}^{h\times w\times n_z}. \quad (3)$$

To generate object patch, we have:

$$\mathbf{p} = G(\mathbf{z}_q) = G(q(E(W(s)))) \quad (4)$$

Although VQGAN generates a series of object patches $\mathcal{P}^c = \{\mathbf{p}_1^c, ..., \mathbf{p}_N^c\}$ related to object category c, the relevant degrees of these object patches to sentences are not consistent. Therefore, we use the DALL-E model to re-rank these patches as follows:

$$\mathcal{P}^c_{re-rank} = TOP_K(\{R(\mathbf{p}_1^c, s), ..., R(\mathbf{p}_N^c, s)\}), \quad (5)$$

where $R(,)$ calculates the relevant degrees and TOP_K function picks the top_k relevant scores of patches for generating synthetic data.

The selected object patches will form a set $\mathcal{P}_s$, and we randomly select object patches from the set $\mathcal{P}_s$ as synthetic images according to the number of objects in each image x_i^s. We randomly pick the patch size in [64, 96, 192] and sample the patches' location in the vertical and horizontal directions of the image. As for the size of the synthetic image, we follow the standard setting in VOC2007 and set it as 600*800.

3.4 Feedback Information Improvement via Transferable Adversarial Attacks

Adversarial examples can fool the model to produce false predictions. Adversarial examples against object detection can attack the detector to make wrong predictions, including misclassifying the detection box, reducing the probability of predicted boxes, and detecting objects in the wrong regions. Research [18, 44] shows that one challenge of attack against object detection is that after successfully attacking the optimal predicted box, the detector still predicts the sub-optimal predicted box near the optimal predicted box. The reason is that the non-maximum suppression algorithm of the detector will suppress the detection boxes at similar positions and output the optimal predicted box with the highest confidence score. Even if the optimal predicted box is successfully attacked,

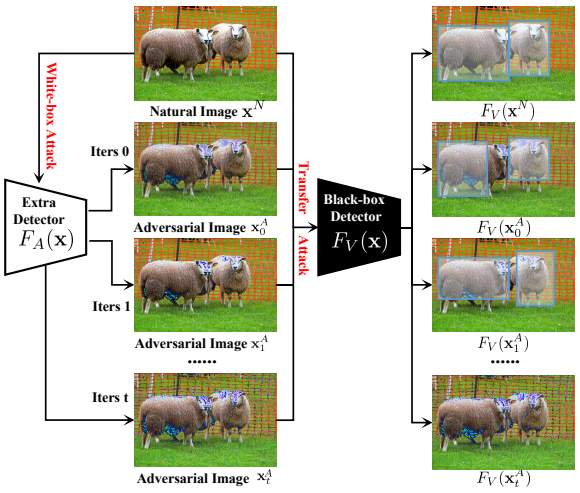


Figure 4: Mining latent information from a black-box model using an iterative transfer-based adversarial attack.

the sub-optimal predicted box will be used as the output result to replace the optimal predicted box.

Although the non-maximum suppression algorithm brings great difficulty to adversarial attacks, in knowledge stealing, we can use this feature to explore the intrinsic knowledge of the object detector fully. Specifically, we construct adversarial examples around the clean image so that the object detector produces different predicted boxes. As shown on the right side in Figure 4, the first row is a clean image $\mathbf{x}^N$, the second row is the adversarial image $\mathbf{x}_0^A$ with random initialization and the remaining images are adversarial samples, and slight differences between similar samples will cause the detector to output different prediction boxes.

The use of adversarial examples can fully exploit the latent knowledge of the victim model, but the choice of the generation method needs to consider two factors: Firstly, the generation of the adversarial example should minimize the access to the victim model. In real-world settings, a large number of queries for similar images in a short period of time will trigger the self-protection mechanism of APIs. Secondly, the goal of adversarial examples is to produce predicted boxes that are different from the optimal predicted box rather than reducing the objectness so that the detector has no output. Based on the above considerations, we choose to use a transfer-based adversarial attack method. Attacking the additional detector $F_A(\mathbf{x})$ by a small number of iterations to generate adversarial examples transfer attacks the victim detector $F_V(\mathbf{x})$, which does not impose a huge query burden on the victim detector. The objective function as follows:

$$\begin{aligned} &\arg\max_{\mathbf{x}_t} l(F_A(\mathbf{x}), F_A(\mathbf{x}_t)) \\ &s.t. ||\mathbf{x}_t - \mathbf{x}||_\infty \leq \epsilon, \mathbf{x}_t = \mathbf{x} + \delta_t, \mathbf{x} \in [0,1]_d \end{aligned} \tag{6}$$

where t denotes the number of iterations. l represents the detector's losses, which are relevant to the structure of the detector. d denotes the dimension of the digital image. ϵ controls the magnitude of the adversarial perturbation δ. The initial perturbation δ samples uniformly in $[-\epsilon, \epsilon]$.

Any transfer-based adversarial attacks, such as DI-FGSM [45] and TI-FGSM [7], can optimize the Equation 6. The choice of this extra detector F_A has special limits, which only require the F_A can detect objects in most of the data in natural datasets X_N. Within a certain number of iterations, with the increase of the number of iterations, the adversarial attack of the adversarial sample against the victim will gradually increase. Victim detectors predict different predicted boxes than clean pictures, which directly increases the sufficiency of model feedback. As a way to augment the training data in the problem domain, iterative transfer-based adversarial attacks increase the sufficiency of the training data.

Algorithm 1: Two-atage Training The Imitated Detector

Input: Victim model F_V, Extra detector F_A, Synthetic model F_G, Imitated detector F_I, Natural images X^N, J Simple text sentences $\{s_1, ..., s_J\}$, number of iterations T in transfer-based attack, number of iterations in the first stage E_1, number of iterations in the second stage E_2, number of samples i in a batch, learning rate η

Output: Parameters of the imitated detector θ_I

Initialize θ_I randomly, $t = 0$, $e_1 = 0$, $e_2 = 0$
Generating synthetic data $X^S = F_G(s_1, ..., s_J)$
while $t > T$ **do**
  Implement the transfer-based attack to optimize equation (6) in natural images X^N
  Generating adversarial data X_t^A
  t=t+1
end
training images in the first stage $X_1 = \{X_T^A, X_0^A, X^S\}$
training images in the second stage $X_2 = X^N$
while $e_1 > E_1$ **do**
  Sample batch of training images $\{\mathbf{x_1}, ..., \mathbf{x_i}\} \sim X_1$
  Computer the loss of model stealing $\mathcal{L}_{KS}$ in equation (2)
  Scheduling the learning rate based on the number of iterations η
  $\theta_I = \theta_I - \eta \nabla_{\theta_I} \mathcal{L}_{KS}$
  $e_1 = e_1 + 1$
end
while $e_2 > E_2$ **do**
  Sample batch of training images $\{\mathbf{x_1}, ..., \mathbf{x_i}\} \sim X_2$
  Computer the loss of model stealing $\mathcal{L}_{KS}$ in equation (2)
  Scheduling the learning rate based on the number of iterations η
  $\theta_I = \theta_I - \eta \nabla_{\theta_I} \mathcal{L}_{KS}$
  $e_2 = e_2 + 1$
end

3.5 Two-stage Training Strategy

We construct three data types in the problem domain, including natural data, adversarial data, and synthetic data, for training imitated

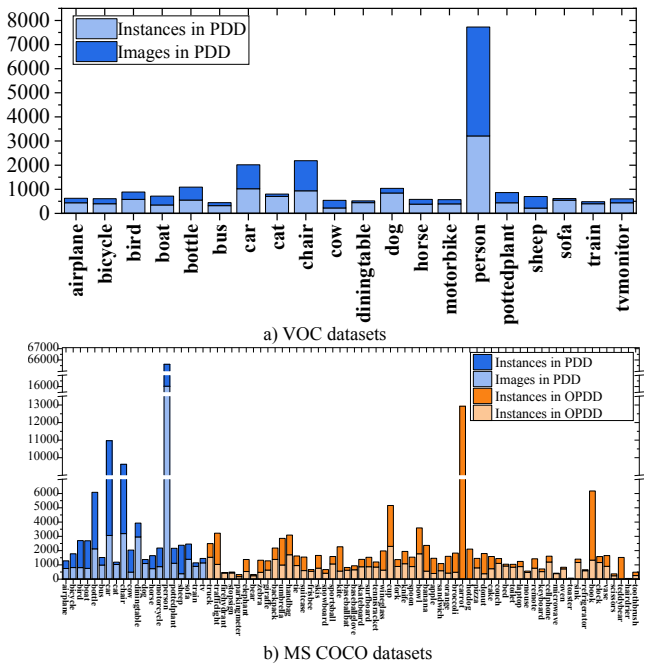


Figure 5: High distribution shifts between ODD, PDD, and OPDD.

models. Since adversarial data [16] and synthetic data are different from natural image distributions, in contrast, the distribution of natural data is closer to the distribution of the original domain data. Therefore, during the training process, we adopt a two-stage training strategy. Specifically, we first use adversarial and synthetic data to query the victim model's APIs to obtain pseudo-labels to train the imitated model. Then, we fine-tune the imitated model on the clean image and their pseudo-labels. The overall training process of the imitated detector can be found in Algorithm 1.

4 EXPERIMENTS

In this section, we first provide the experimental settings, then elaborate on our main object detector stealing experiments, ablation studies, and discussions.

4.1 Settings

Datasets. We adopt the commonly-used and standard object detection datasets including VOC2007 [10], VOC2012 [9], and MS COCO datasets [20]. In particular, VOC datasets contain 20 object categories, while MS COCO has 80 categories, including all categories of the VOC dataset. We define the 20 categories of the VOC dataset as object categories and use about 8000 images of all VOC2007 and VOC2012 training data as the *original domain data* (ODD) to train the victim model. We define 20 VOC categories in MS COCO as *problem domain data* (PDD), and the remaining 60 categories as *out of the problem domain data* (OPDD). Since it is not practical to collect high-quality data in real scenarios, we select 25% of PDD and OPDD in MS COCO as natural datasets, consisting of about 4500 images. We use the test datasets of VOC2007 as the evaluation dataset, including 4952 images. The scene categories used to compose simple text sentences come from the UC Merced dataset [47], which includes 11 common scene categories.

Object Detectors. For our main model stealing experiment, since the anchor-free detectors are gradually becoming a popular trend and do not need to tune the number of parameters for the dataset, we hereby choose the Retinanet detector [19] based on ResNext-101 [46] as the *victim detector* and use the Retinanet detector based on ResNet-50 as the imitated detector. Moreover, we also study the possibilities of performing model extraction attacks with different victim/imitated model architectures, where we further choose FCOS [33], Free anchor [48], FSAF [49], and guided anchoring [46] in Section 4.4.1. For the extra detector that produces predicted boxes on natural data based on the transfer-based adversarial attack, we use the publicly available Faster R-CNN model trained on the MS COCO dataset[1].

Evaluation metrics. Since our goal is to steal the knowledge of object detectors, we therefore evaluate the performance of imitated models using MS COCO's evaluation criteria [20]. Average precision is reported in two forms: absolute (x) or relative (x×) to the victim detector $F_V(\mathbf{x})$. For all the metrics mentioned above, the higher, the better stealing performance.

Implementation details. All of our experiments are conducted based on 8 V100 GPUs. For the *training of our framework*, we first iterate for 12 epochs in the first stage and 6 epochs in the second stage. The initial learning rate is 0.01 and decays to 0.001 after 8 epochs. Except for the two-stage training strategy, the imitated models are iteratively trained 12 epochs on the corresponding training set. For the *transfer-based adversarial attack*, we use M-DI-FGSM attacks [45] with the perturbation budget $\delta = 0.05$. For the *synthetic model*, we use pre-trained models based on minGPT [5], VQGAN [8], and DALL-E [29].[2]

4.2 Comparison with Other Baselines

In this part, we compare our Imitated Detector framework with other model extraction attack methods. Since there exist no studies on knowledge stealing for object detectors, we therefore introduce several knowledge stealing baselines from image classification and adapt them to the object detection task. In classification, the training dataset of Knockoff achieves the best results on 100% ODD, And Copycat exploits different distributions of the training datasets to improve the imitated model. Specifically, we adopt the commonly-used Knockoff, Copycat, and our proposed method. Knockoff uses a random strategy to sample original domain data (ODD) due to the logistic score of the model feedback; Copycat steals the knowledge from the problem domain data (PDD) and out of the problem domain data (OPDD). Our proposed method includes constructing adversarial and synthetic data and a two-stage training strategy.

In object detection, we divide the dataset into ODD, PDD, and OPDD. As shown in figure 5, we can observe significant distribution differences between them. Following the setting of knowledge stealing in image classification [6, 27], we conduct comparisons

[1]The codes and weights are publicly available at https://github.com/open-mmlab/mmdetection

[2]The codes and weights are publicly available at https://github.com/kakaobrain/minDALL-E.

Table 1: Knowledge stealing results in different training datasets settings. Our approach outperforms others by large margins.

Dataset	Distribution(Rate)	Model	Params(M)	mAP	mAP_50	mAP_75	mAP_s	mAP_m	mAP_l
X^N	ODD(100%)	F_V	722	0.514	0.798	0.549	0.234	0.362	0.569
X^N	ODD(100%)	F_I	280	**0.386(0.75×)**	**0.680**	**0.387**	**0.138**	**0.262**	**0.430**
X^N	PDD+OPDD(25%)	Copycat	280	0.217(0.42×)	0.428	0.195	0.096	0.154	0.245
X^N	PDD(25%)	Copycat	280	0.225(0.42×)	0.433	0.207	0.172	0.180	0.246
X^N	ODD(50%)	Knockoff	280	0.253(0.49×)	0.491	0.230	0.090	0.175	0.290
$\{X^A, X^S\}, X^N$	PDD(25%)	**Ours**	280	**0.358(0.70×)**	**0.635**	**0.355**	**0.162**	**0.273**	**0.393**

Table 2: Ablation studies on the effects of different training data groups for the imitated detector.

Method	mAP	mAP_50	mAP_75	mAP_s	mAP_m	mAP_l
X^N	0.211	0.395	0.203	0.108	0.166	0.232
X_0^A	0.170	0.326	0.159	0.098	0.136	0.183
X_4^A	0.193	0.369	0.181	0.142	0.155	0.207
$\{X_0^A, X_4^A\}$	0.290	0.529	0.281	0.147	0.214	0.320
$\{X_0^A, X_4^A, X^N\}$	0.321	0.595	0.307	0.149	0.242	0.353
$\{X^S, X^N\}$	0.324	0.598	0.314	0.148	0.245	0.355
$\{X_0^A, X_4^A, X^S, X^N\}$	0.351	0.628	0.350	0.160	0.264	0.390
$\{X_0^A, X_4^A, X^S\}, X^N$	**0.358**	**0.635**	**0.355**	**0.162**	**0.273**	**0.393**

under different training data combination scenarios, which could help us to better explore the superiority of our framework in the more practical and challenging scenarios. In detail, the Knockoff randomly samples 50% ODD data due to lacking the logistic score; Copycat conducts the model stealing in 25% PDD or 25% OPDD. For our framework, we use 25% PDD data for training. Moreover, we also report the results of an imitated model on 100% ODD data through teacher-student training as F_I.

The results in Table 1 demonstrate that **our Imitated Detector could steal the knowledge of the victim model on the small amount of data without knowing the training data of the victim model, even achieving comparable performance to F_I that directly access the victim's training dataset**. Compared to Knockoff and Copycat, we can draw multiple conclusions: (1) our framework could outperform other methods by large margins using the same scale of training data (+50% on average, *e.g.*, **0.358** *v.s* 0.225 and 0.253); (2) comparing with the result of F_I that uses 100% ODD that represents the upper bound of knowledge that obtained by the imitated model in the the victim model's dataset, our proposed method achieves similar results in different metrics (xxx v.s. xxx), which showing the effective knowledge stealing ability of our method; (3) collecting domain-specific data is beneficial for knowledge stealing (0.217 v.s. 0.225 for Copycat with additional OPDD); (4) note that the imitated model has about 1/3 the number of parameters of the victim model(280M v.s. 722M), which demonstrate that we can use a small model to steal the knowledge of a large model.

4.3 Ablation Studies

In this part, we conduct ablation studies to investigate the effectiveness of the two core parts of our framework (*i.e.*, synthetic data generation and iterative adversarial attacks in the problem domain) and our two-stage training strategy.

Specifically, we conduct several pairs of experiments as follows: (1) training on adversarial data at different iteration steps, *i.e.*, 0-*th* iteration X_0^A, 4-*th* iteration X_4^A, and the mixture $\{X_0^A, X_4^A\}$; (2) training on natural and adversarial data, *i.e.*, $\{X_0^A, X_4^A, X_N\}$; (3) training on natural and synthetic data, *i.e.*, $\{X^S, X_N\}$; (4) training on natural, adversarial, and synthetic data, *i.e.*, $\{X_0^A, X_4^A, X^S, X_N\}$; and (5) training on natural, adversarial, and synthetic data in a two stage manner, *i.e.*, $\{X_0^A, X_4^A, X^S\}, X_N$. For all experiments from (1) to (4), the models are trained in a single-stage manner.

As shown in Table 2, we can draw several observations regarding the performance of the imitated model Retinanet detector in the problem domain as follows (1) comparing to model trained solely on natural data X_N, intimated detectors trained on additional adversarial data $\{X_0^A, X_4^A, X_N\}$ or synthetic data $\{X^S, X_N\}$ respectively could all improve the model stealing performance; (2) by training on all data (*i.e.*, $\{X_0^A, X_4^A, X^S, X_N\}$), our intimated detector could further be improved compared to using only one extra type of data; (3) two-stage training strategy results in better performance than the one-stage manner; (4) interestingly, we find that the results of natural data X_N are higher than that of adversarial data with one specific iteration steps (*i.e.*, X_0^A or X_4^A). We conjecture that this is due to the distribution of natural data in PDD and ODD being closer than that of adversarial data in PDD and natural data in ODD. However, training on the mixture of adversarial data $\{X_0^A, X_4^A\}$ significantly increases the knowledge stealing performance, which we believe might be that adversarial examples with different iterations provide different latent information of the victim model.

To sum up, our ablation studies demonstrate that our synthetic data generation, iterative adversarial attacks, and the two-stage training strategy could effectively improve the model stealing performance.

4.4 Discussions and Analyses

4.4.1 ***Stealing with Different Victim/Imitated Architectures.*** It is apparent that using a student model that is structurally similar to the teacher model would result in a higher success rate. However, in a practical knowledge stealing scenario, the attacker often does not know the victim model's specific structure. Therefore, in this part, we study the possibilities of detector knowledge stealing between the victim and imitated models with different architectures. Specifically, we use Retinanet for the victim model, and adopt FCOS, Anchor Free, Guided Anchoring, and FSAF as the imitated model

Table 3: Experiments of different victim/imitated detectors architectures.

Detector	mAP	mAP_50	mAP_75	mAP_s	mAP_m	mAP_l
F_V	0.514	0.798	0.549	0.234	0.362	0.569
Retinanet	0.358(0.70×)	0.635	0.355	0.162	0.273	0.393
FCOS	0.325(0.63×)	0.601	0.315	0.155	0.260	0.360
Anchor Free	**0.382(0.74×)**	**0.644**	**0.399**	**0.200**	**0.279**	**0.423**
Guided Anchoring	0.355(0.69×)	0.645	0.354	0.144	0.259	0.393
FSAF	0.372(0.72×)	0.634	0.385	0.147	0.273	0.413

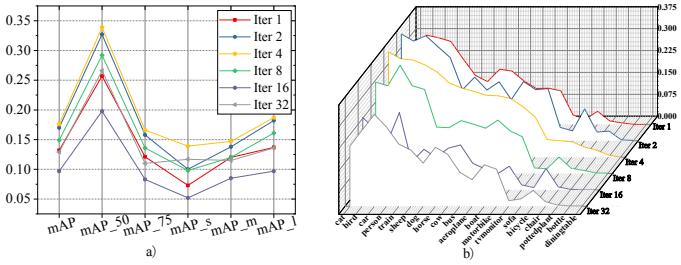


Figure 6: Effects of adversarial data on the imitated detector. a) represents results of the imitated detector on different iterations. b) shows results on different iterations for every category.

backbone. These imitated models (*i.e.*, FCOS, Anchor Free, Guided Anchoring, and FSAF) differ significantly from Retinanet in both loss function design and implementation details.

The experimental results in Table 3 show that **under the different architectures scenario, the adversary could steal victim model knowledge to a certain extent as well**. For example, the worst-performing FCOS model also achieved a 63% correlation with the victim model. Then, different model structures also significantly impact the stealing effect. However, we believe that this may be a performance difference caused by the structural difference of the model itself. The Anchor Free model outperforms the FCOS model under suitable training parameters. Therefore, using a detector with better performance as an imitated model is also an option.

4.4.2 ***Analysis of Adversarial Attacks.*** The strength of the adversarial attack will directly affect the feedback information of the victim model. Therefore, we further provide a detailed analysis of adversarial attacks on the effectiveness of our framework.

An overly strong transfer attack will cause the victim model to produce no output; and a weak transfer attack will make the model produce predictions indistinguishable from the natural data and cannot effectively exploit model information. The **number of iterations** is an important factor affecting the attacking performance of the transfer-based attack. The more iterative attacks within a certain range, the stronger the transferability of attacks. Here, we choose 1, 2, 4, 8, 16, and 32 as the iterations of the transfer-based attack, respectively. Figure 6 shows the training of the imitated model on adversarial data generated at different iterations. As the number of iterations increases, the imitated model peaks and then decreases. More precisely, we also show the training results of the imitated detector with different iterations for each category in Figure 6, and we can observe that the training result of the imitated

Table 4: Experiments of different adversarial attacks.

Method	mAP	mAP_50	mAP_75	mAP_s	mAP_m	mAP_l
DI-FGSM	0.098	0.198	0.085	0.085	0.086	0.104
M-DI-FGSM	**0.193**	**0.369**	**0.181**	**0.142**	**0.155**	**0.207**
TI-FGSM	0.140	0.275	0.127	0.09	0.114	0.151
M-TI-FGSM	0.148	0.293	0.135	0.072	0.115	0.162
TI-DI-FGSM	0.159	0.313	0.145	0.091	0.13	0.172
M-DI-TI-FGSM	0.149	0.295	0.134	0.078	0.117	0.161

model is the best with iteration number 4. It can also be seen that the knowledge stealing difficulty varies greatly among categories, such as cats and dining table, which may be due to adversarial attacks that cause different data distributions. Designing a transfer attack method generateing balance categories is also a direction. Overall, we choose the iteration number of the iterative attack to be 4 in our main experiment.

Meanwhile, since our proposed objective function of the transfer-based adversarial attack is suitable for a variety of methods, it is also worth studying the effectiveness of **different adversarial attack methods**. In particular, we select 4 representative attack algorithms, including DI-FGSM, TI-FGSM, and their combined versions with momentum attack (M-DI-FGSM, M-TI-FGSM). Table 4 shows the training results of imitated models on datasets generated by different transfer-based adversarial attacks, where the number of iterative attacks is set to 4. The M-DI-FGSM algorithm achieves the best results, which may be that the adversarial data generated by this algorithm has better transferability and a larger gap with clean images, which can mine potential information about victim models.

5 CONCLUSION

In this paper, we reveal that black-box victim object detectors can be easily stolen without knowing the structures of detectors and training datasets for the first time, which may threaten intellectual property and model safety. Different from classification, stealing knowledge against object detection faces the challenge of the efficiency and sufficiency for constructing the training dataset. To settle the above challenge, we propose a teacher-student framework to train the imitated detector by text-image generation and transferable adversarial attacks to construct adversarial data and synthetic data from the natural collecting data. Experiments show that we can steal the knowledge of victim detectors using imitated detectors with different architectures. In the case of different training data distribution from the victim detector and limited training data size, the victim detector can still be effectively replicated and outperform other model stealing methods, which means that imitated detectors can steal the knowledge of black-box victim detectors in scenarios close to the real world.

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